PLM — How to Create BOM Revisions

Engineering Change Order (ECO): modify an existing BOM

Source: Odoo Tour Recorder export — `Tour/PLM_1 - How to Create BOM Revisions.json`

Scenario: An existing Bill of Materials is modified through an **Engineering Change Order**: a new BOM version is created as a copy, a component quantity is changed and a component is added, then the ECO is validated, approved and applied so that the new version becomes the active BOM.

Every screenshot below is a real capture from a single, clean, passing automated run — taken *before* each step's action, with the element to use framed in red.

1. Overview

This manual explains how to create a **BOM revision** in the **Product Lifecycle Management (PLM)** app. PLM is used to create a new product, a new BOM and new BOM versions. A revision is made through an ECO: **Start Revision** copies the final BOM, the changes are made in the copy, and **Apply Changes** activates the new version.

ECO stages covered

New | In Progress | Validated | Effective

Before you start

- You have access to the PLM (Engineering Changes) and Manufacturing apps and are allowed to approve ECOs.
- The active company (top bar) is **Jinasena Agricultural Machinery (Pvt) Ltd.**
- The product already exists, is an **Approved Item** and has a Bill of Materials.
- Naming standard: the product code to be revised starts with **M-** and ends with **-V** followed by the next sequence number (e.g. `M-JMC10D-V1.0`).

Example data used in this manual

Field	Value
ECO name	M-JMC10D-V1.0
Product	[JMC10D] Multi Chopper with 10Hp. Engine (the screenshots show a [TEST] copy of it)
Quantity change	First component: 20 → 18
Added component	[02BB 023] BALL BEARING 6202-2RS

How to read each step

Navigation	Where you are in Odoo (app › menu › screen / dialog).
Action	What to click, select, or type.
Description	Why the step is needed and what happens.
Screenshot	The screen you are on *right before* performing the action, with the element to use framed in red.

Step summary

Section	Steps	Purpose
A	1–3	Open the PLM app and start a new Engineering Change Order (ECO).
B	4–7	Enter the ECO name by the company standard, select the product and create the new BOM version.
C	8–13	Inside the BOM copy, change a component quantity and add a new component, then return to the ECO.
D	14–18	Check the BoM Changes, move the ECO to Validated, approve it and apply the changes to activate the revision.
E	19–22	Open the product and its Bill of Materials to confirm the new version is active.

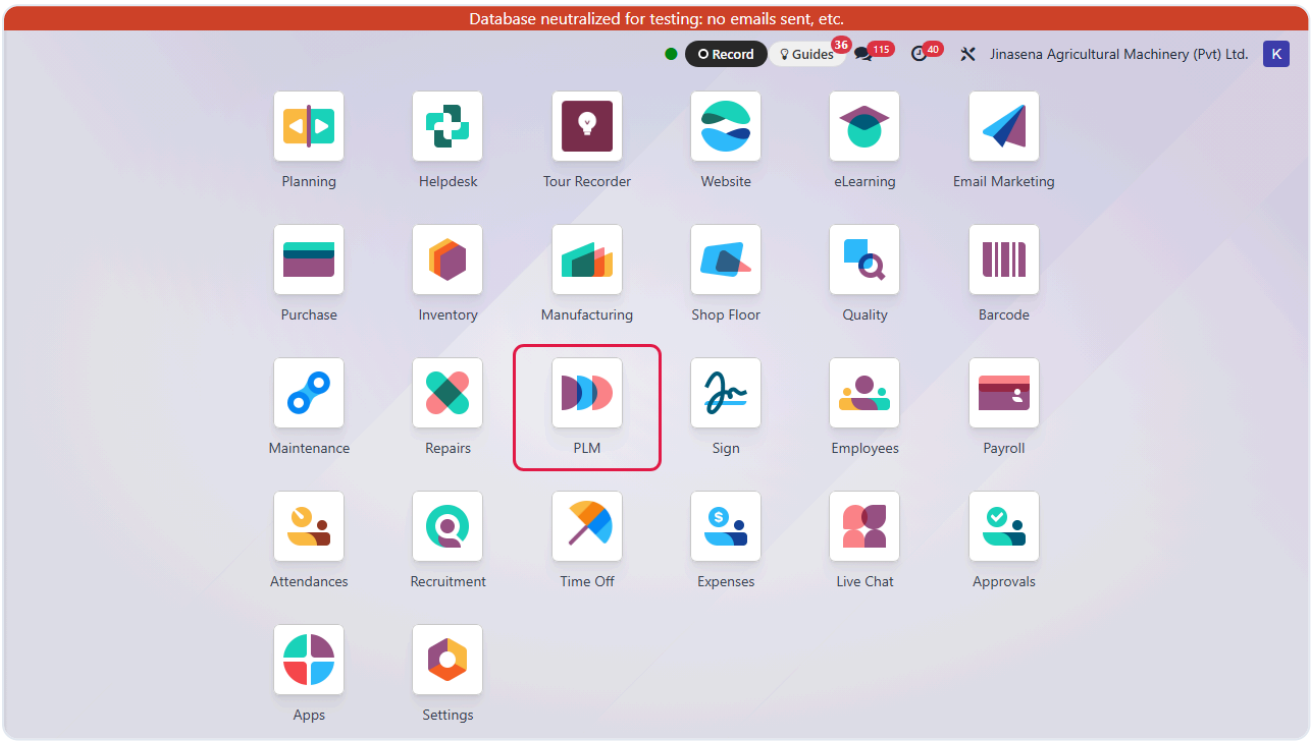
Section A — Open Engineering Changes and create the ECO (Steps 1–3)

1 Open the PLM app

Navigation Odoo Home → Apps

Action **Click the PLM (Product Lifecycle Management) app icon on the home screen.**

Description PLM is used to create a new product, a new BOM and new BOM versions. Opens the PLM Overview dashboard.



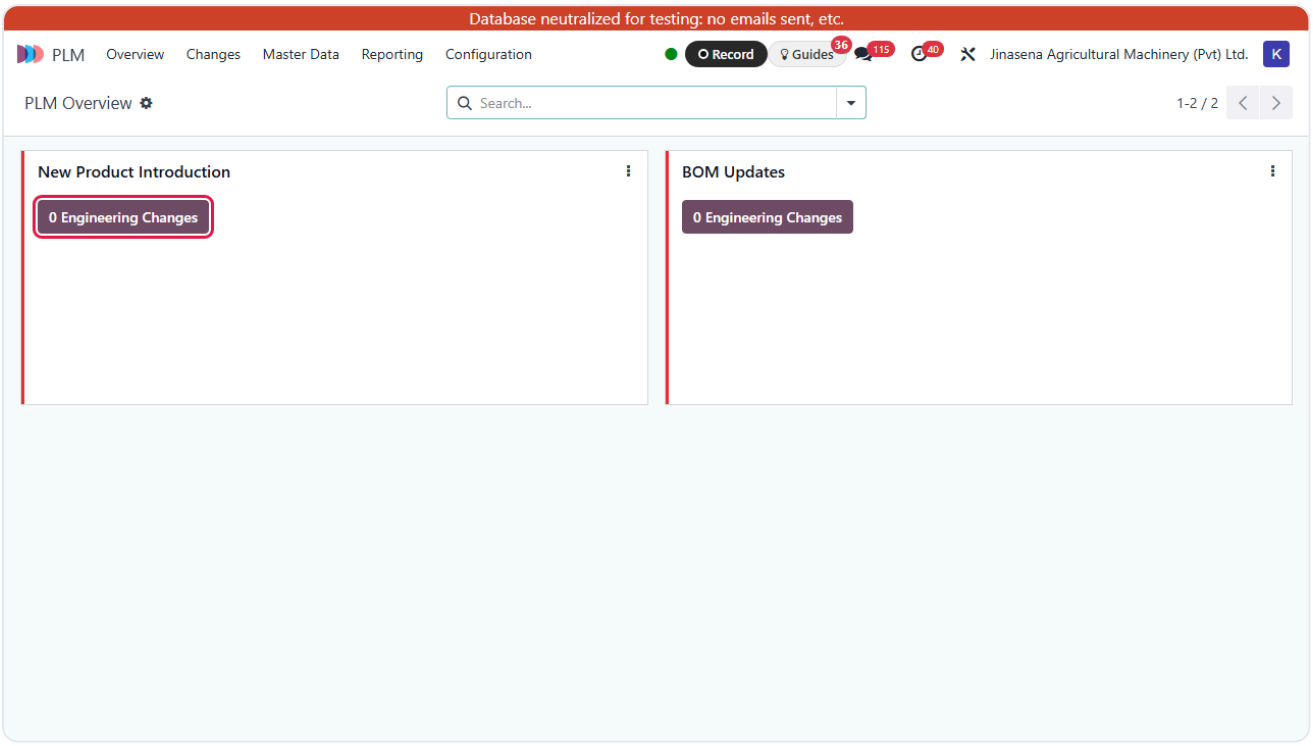
Odoo Apps home screen — PLM app icon

2 Open Engineering Changes

Navigation PLM → Overview

Action **Click the Engineering Changes button on the dashboard card (it shows the number of ECOs).**

Description Click here to start the process. Lists the existing Engineering Change Orders.



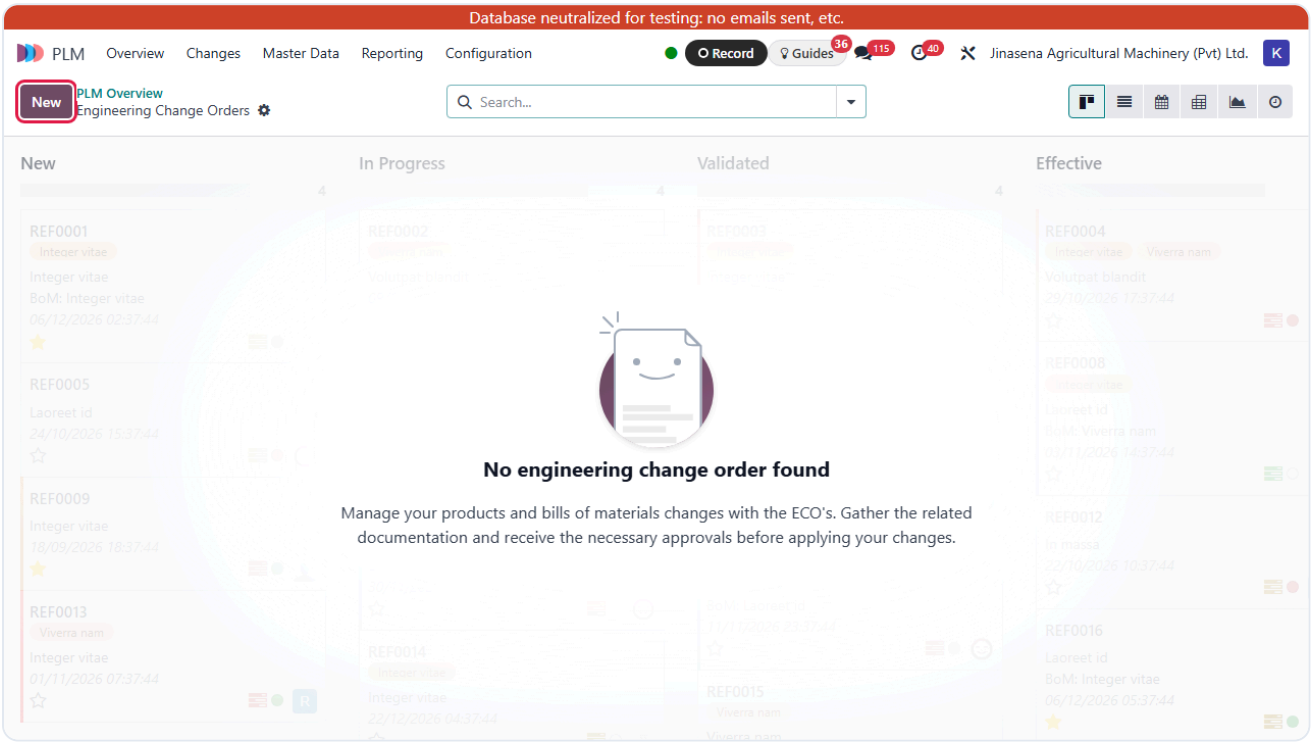
PLM Overview — Engineering Changes button

3 Create a new ECO

Navigation PLM → Engineering Change Orders

Action **Click New.**

Description Opens a blank ECO (Engineering Change Order) form.



Engineering Change Orders list — New button

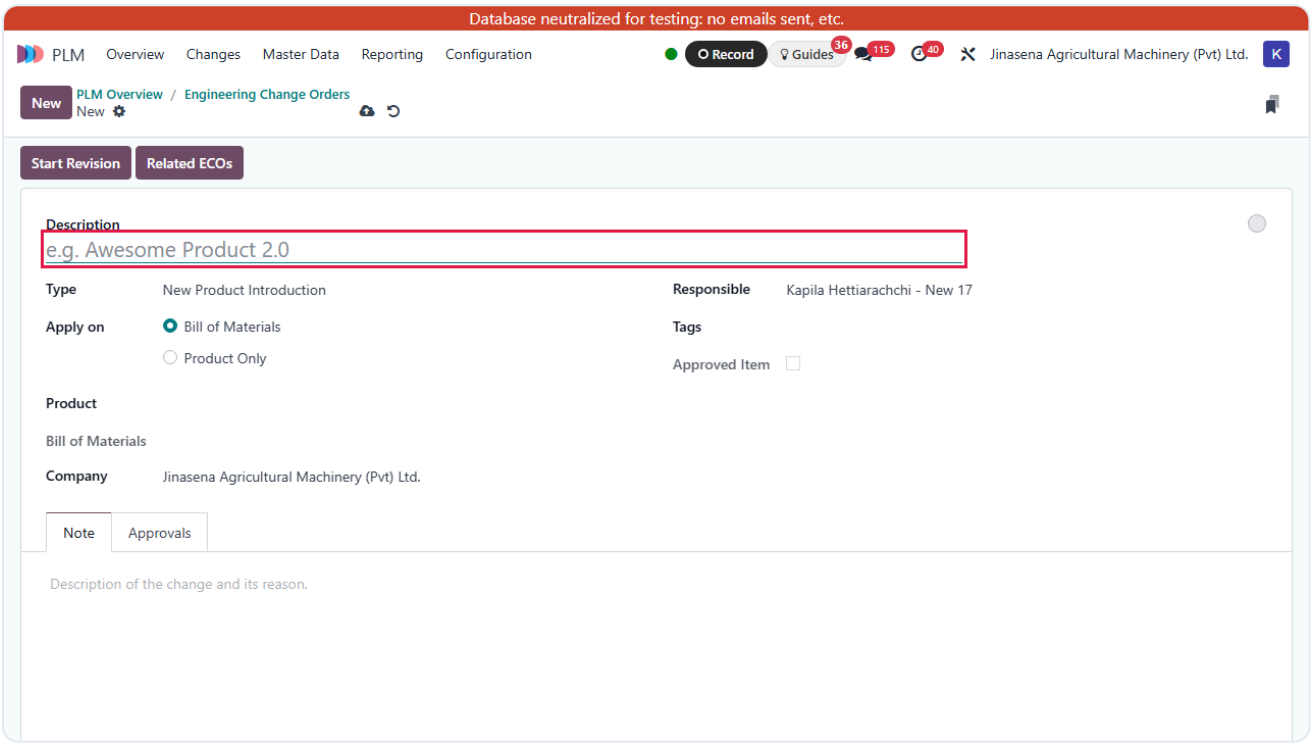
Section B — Name the ECO, choose the product and start the revision (Steps 4–7)

4 Enter the ECO name

Navigation PLM → Engineering Change Orders → New

Action **Click the Description (name) field and type the ECO name, e.g. `M-JMC10D-V1.0`.**

Description According to the company STANDARD, the PRODUCT CODE of the product being revised must start with **M-**, and the next sequence number, preceded by **-V**, is added to the end of the product code (e.g. `M-JMC10D-V1.0`).



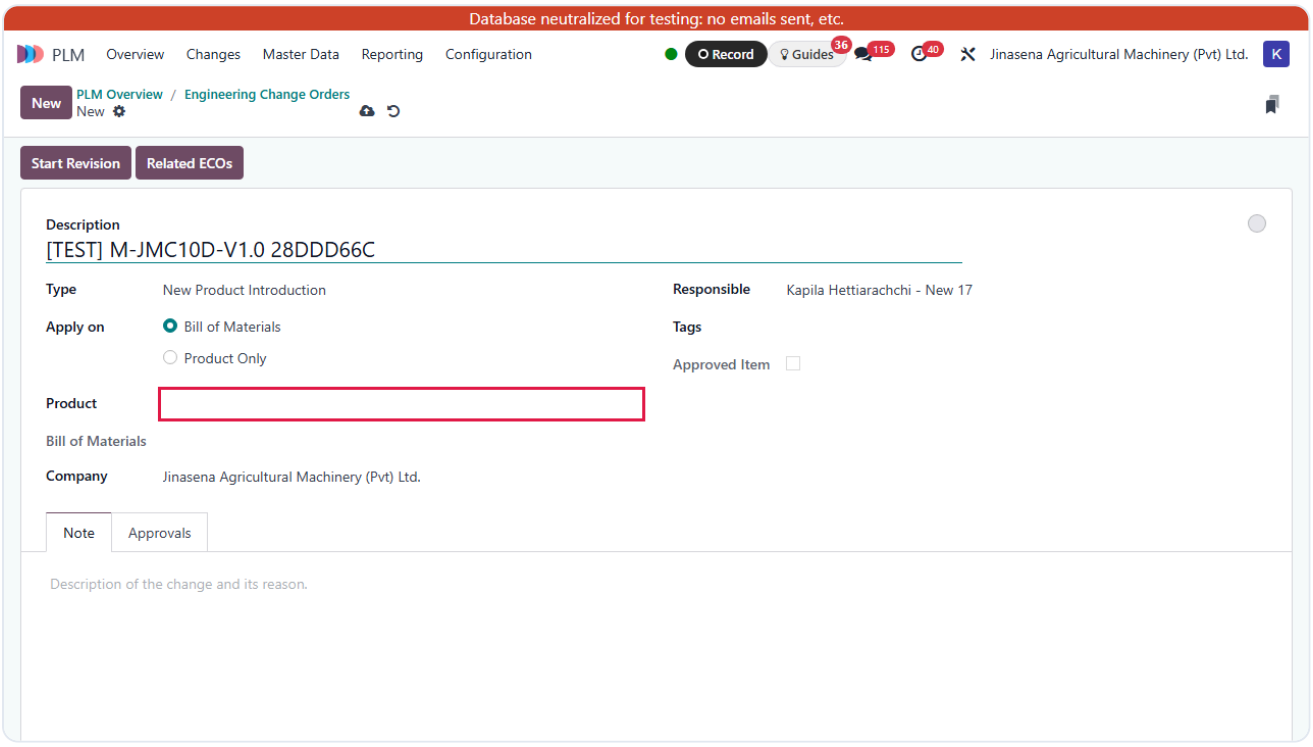
New ECO — Description (name) field

5 Select the product

Navigation PLM → Engineering Change Orders → New

Action **Click the Product field, type the product code (e.g. `JMC10D`) and press Enter / choose the product.**

Description The **Bill of Materials** of the product is filled in automatically. The product must be an *Approved Item*, otherwise the ECO shows the error "The Selected Product Should be Approved to Proceed."



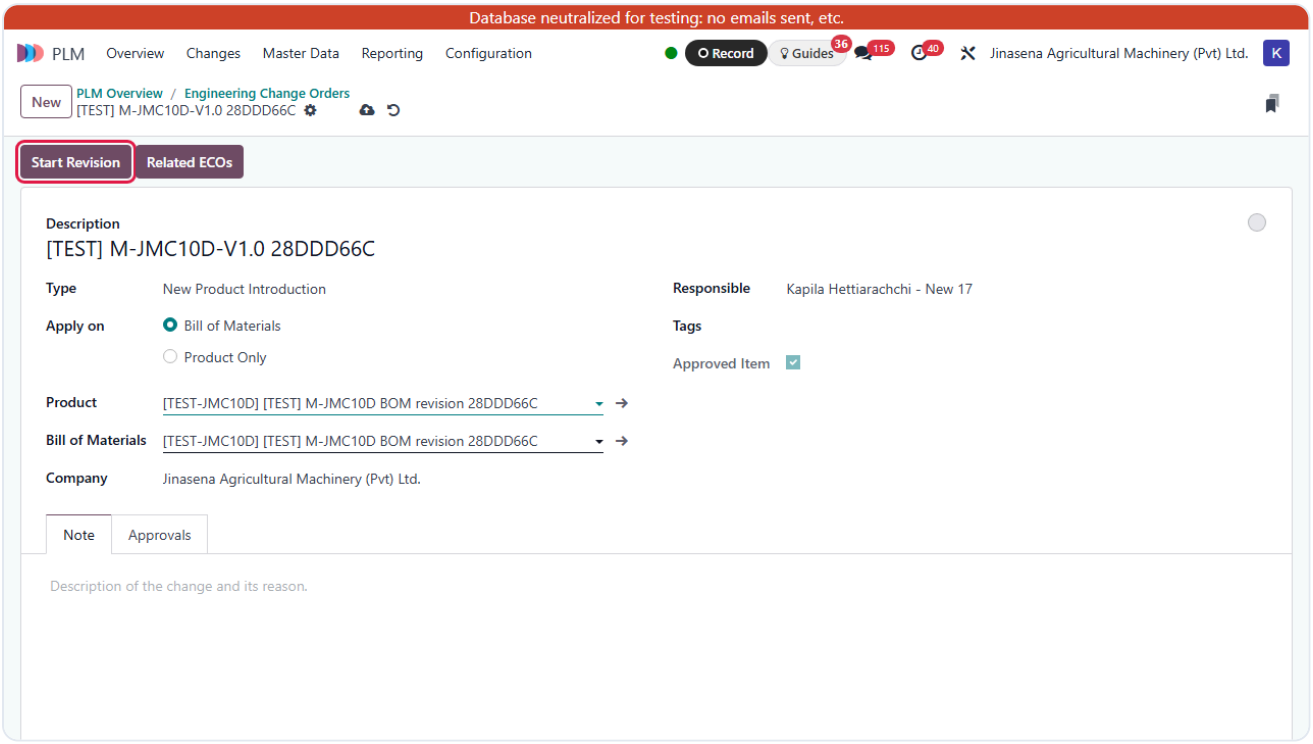
New ECO — Product field

6 Start the revision

Navigation PLM → Engineering Change Orders → ECO

Action **Click Start Revision.**

Description Creates the new BOM version (V2) as a copy of the final BOM and links it to the ECO.



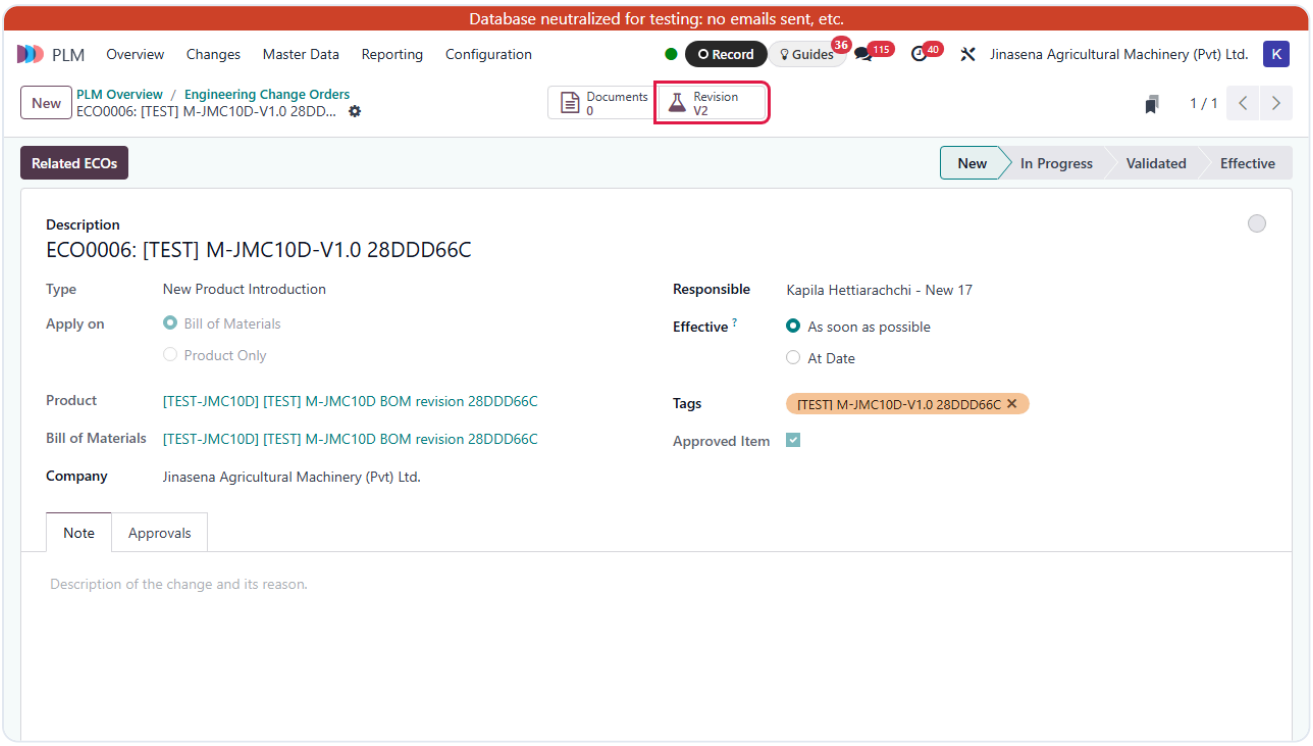
ECO — Start Revision button

7 Open the new BOM version

Navigation PLM → Engineering Change Orders → ECO

Action **Click the Revision V2 button at the top of the ECO.**

Description A new version has been created by making a copy of the final BOM. The changes must be made within this copy. Click here to open it.



ECO — Revision V2 button

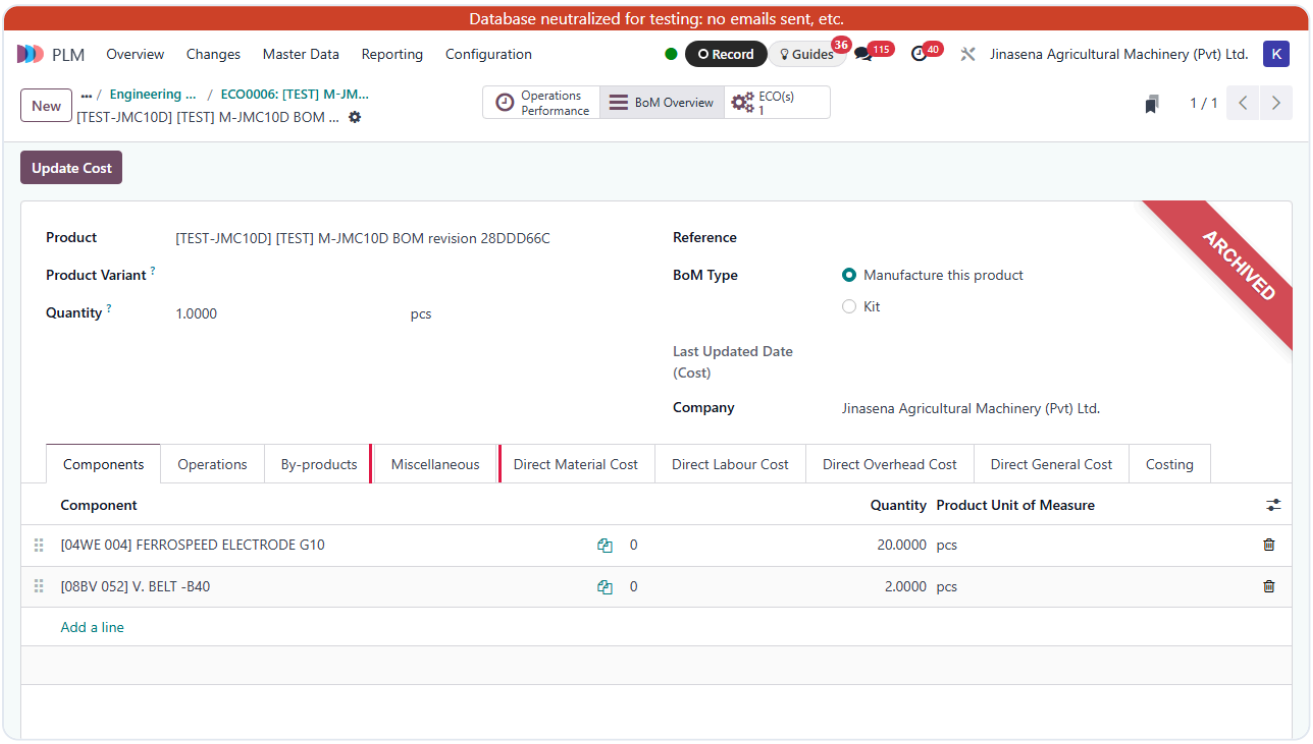
Section C — Change the components of the new BOM version (Steps 8–13)

8 Check that the copy is archived

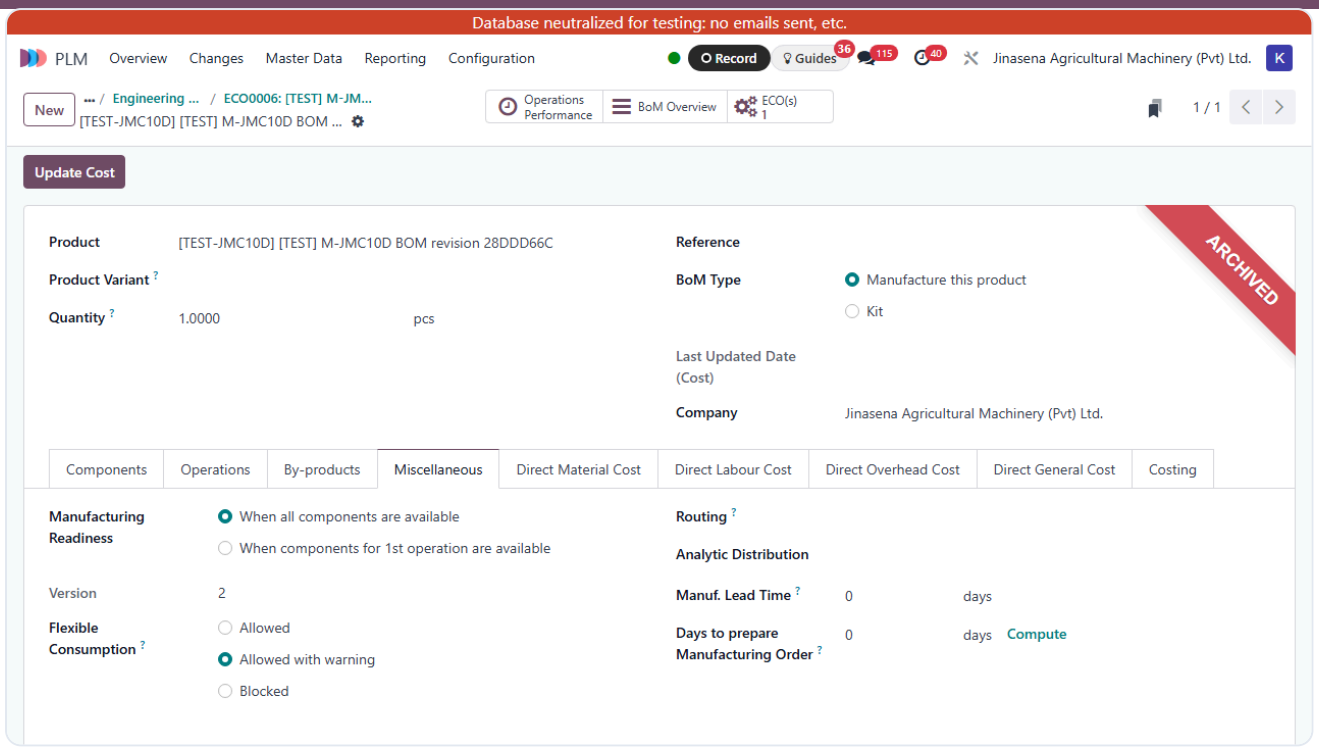
Navigation PLM → ECO → Revision V2 (Bill of Materials)

Action **Click the Miscellaneous tab.**

Description The new BOM version is currently in an **ARCHIVED** state (see the ribbon) and shows its **Version** number. It becomes active only when the ECO is applied.



New BOM version — Miscellaneous tab



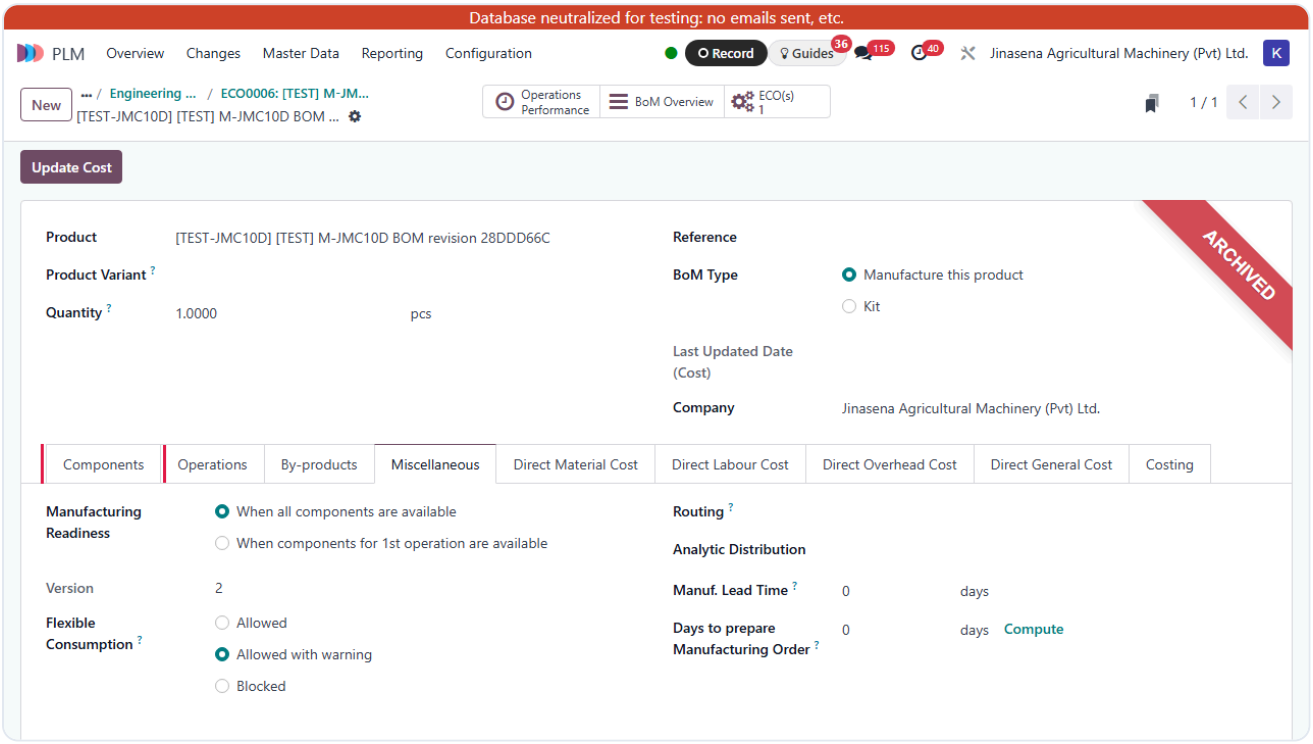
Miscellaneous tab — ARCHIVED ribbon and Version 2

9 Open the Components tab

Navigation PLM → ECO → Revision V2 (Bill of Materials)

Action **Click the Components tab.**

Description View the existing components, or add, delete or modify them.



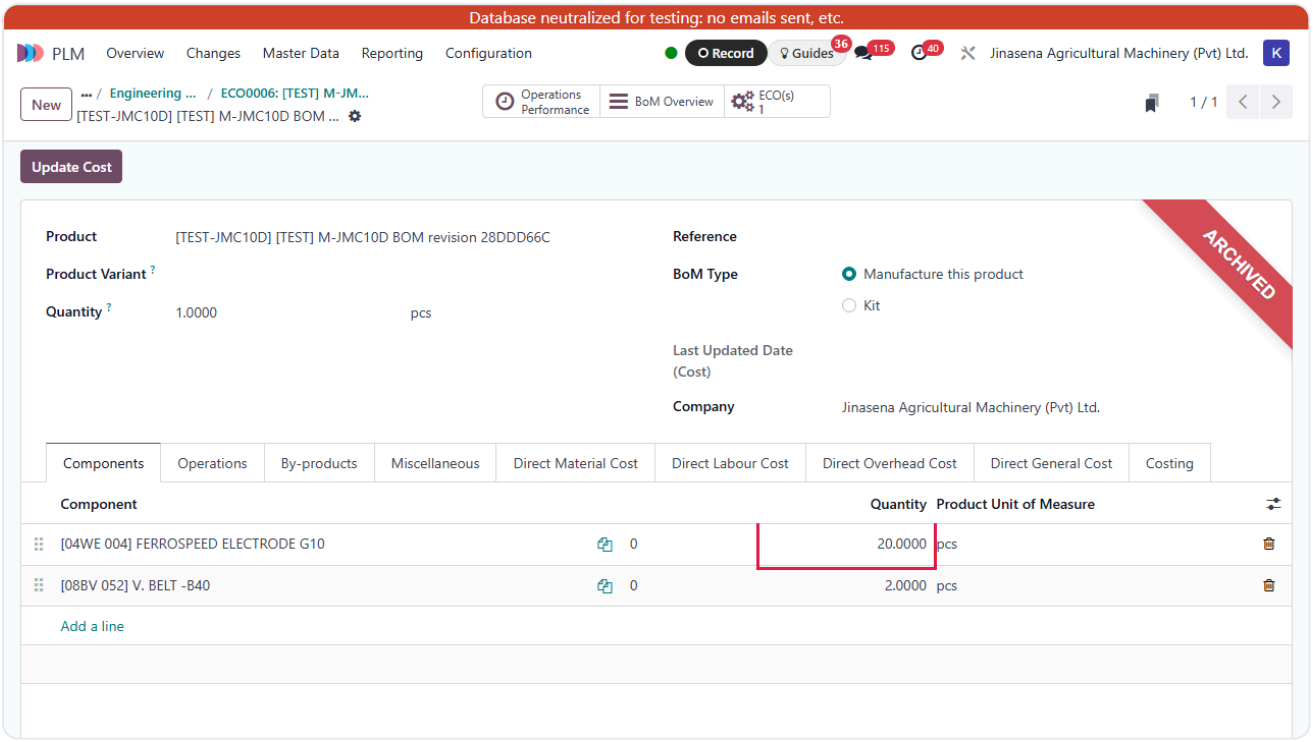
New BOM version — Components tab

10 Change a quantity

Navigation Bill of Materials → Components

Action **Click the Quantity cell of the first line (e.g. 20.0000), type `18` and press Enter.**

Description Changes the quantity of that component in the new version only.



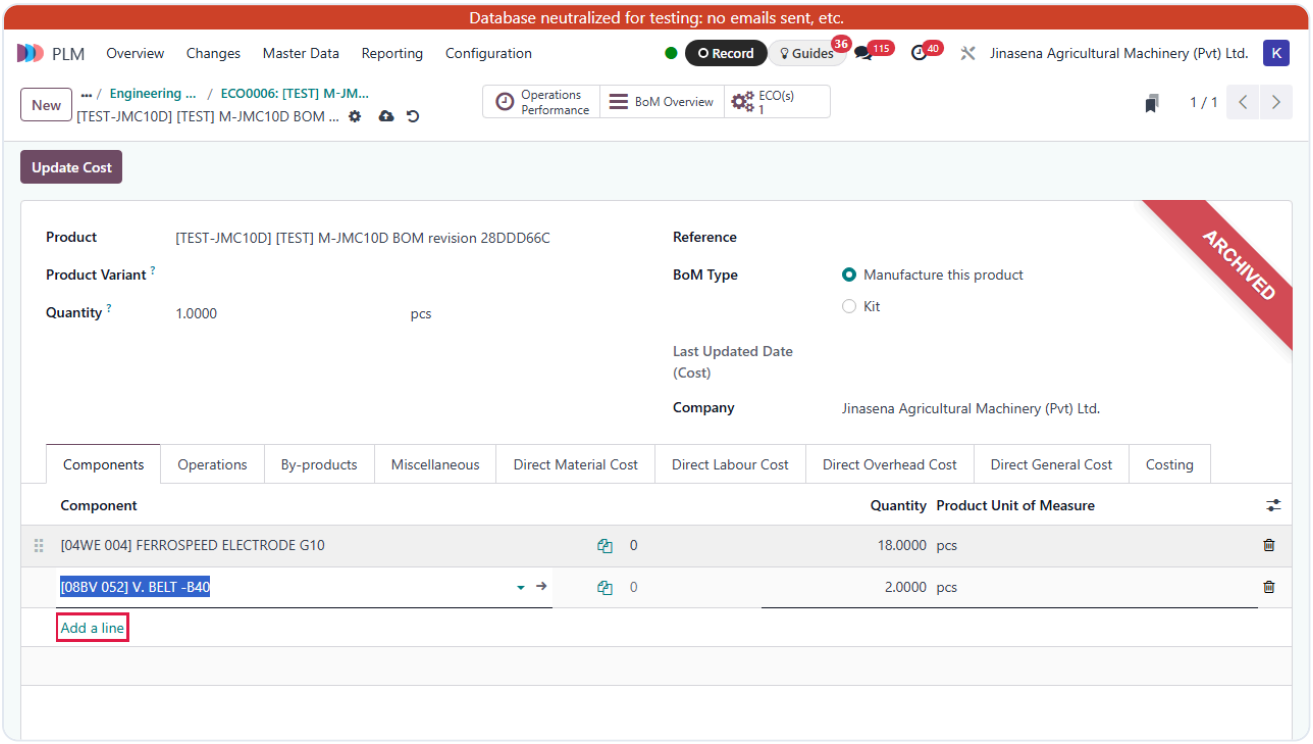
Components — Quantity cell of the first line

11 Add a line

Navigation Bill of Materials → Components

Action **Click Add a line.**

Description Adds a new component row.



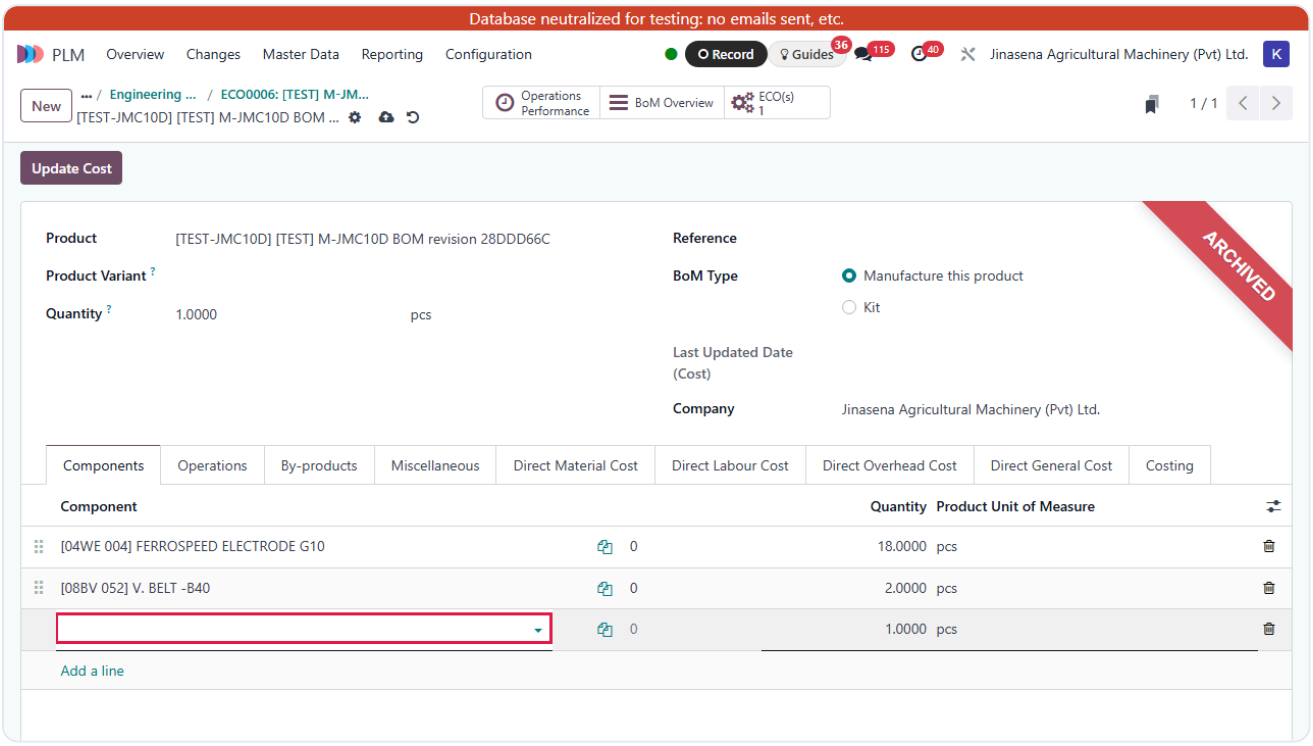
Components — Add a line

12 Add the new component

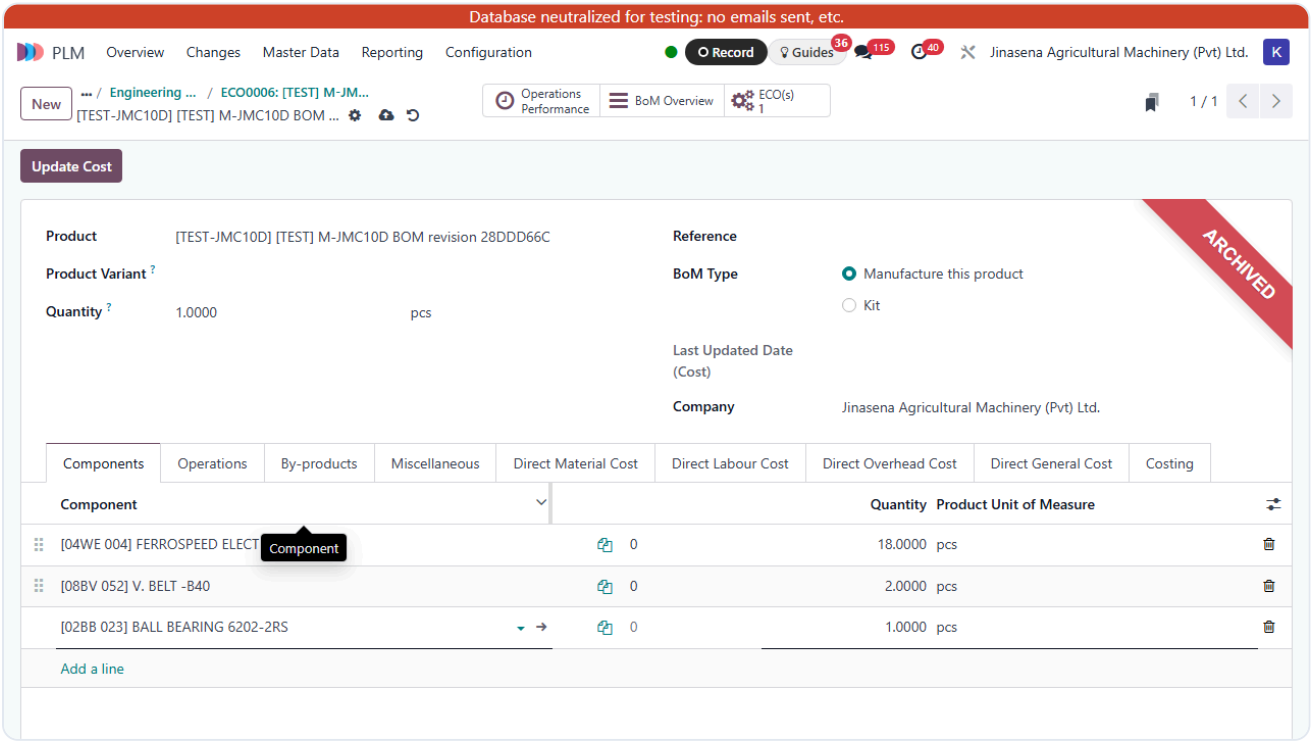
Navigation Bill of Materials → Components

Action **In the Component field type the product code (e.g. `02BB 023`) and press Enter to add it.**

Description The component [02BB 023] BALL BEARING 6202-2RS is added to the new version.



Components — new line, Component field



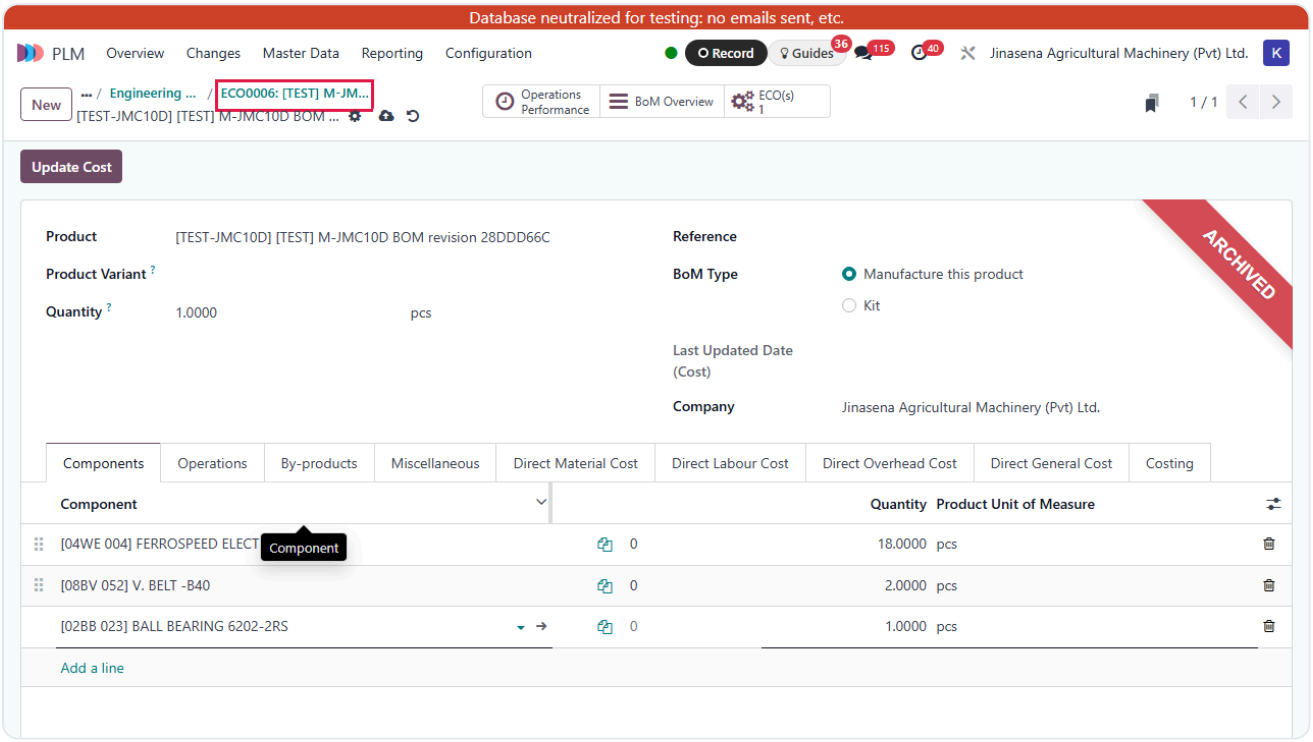
Components — component added

13 Go back to the ECO

Navigation Bill of Materials (breadcrumb)

Action **Click the ECO name in the breadcrumb (top left), e.g. ECO0079: ....**

Description Saves the BOM changes and returns to the ECO.



Bill of Materials — ECO breadcrumb

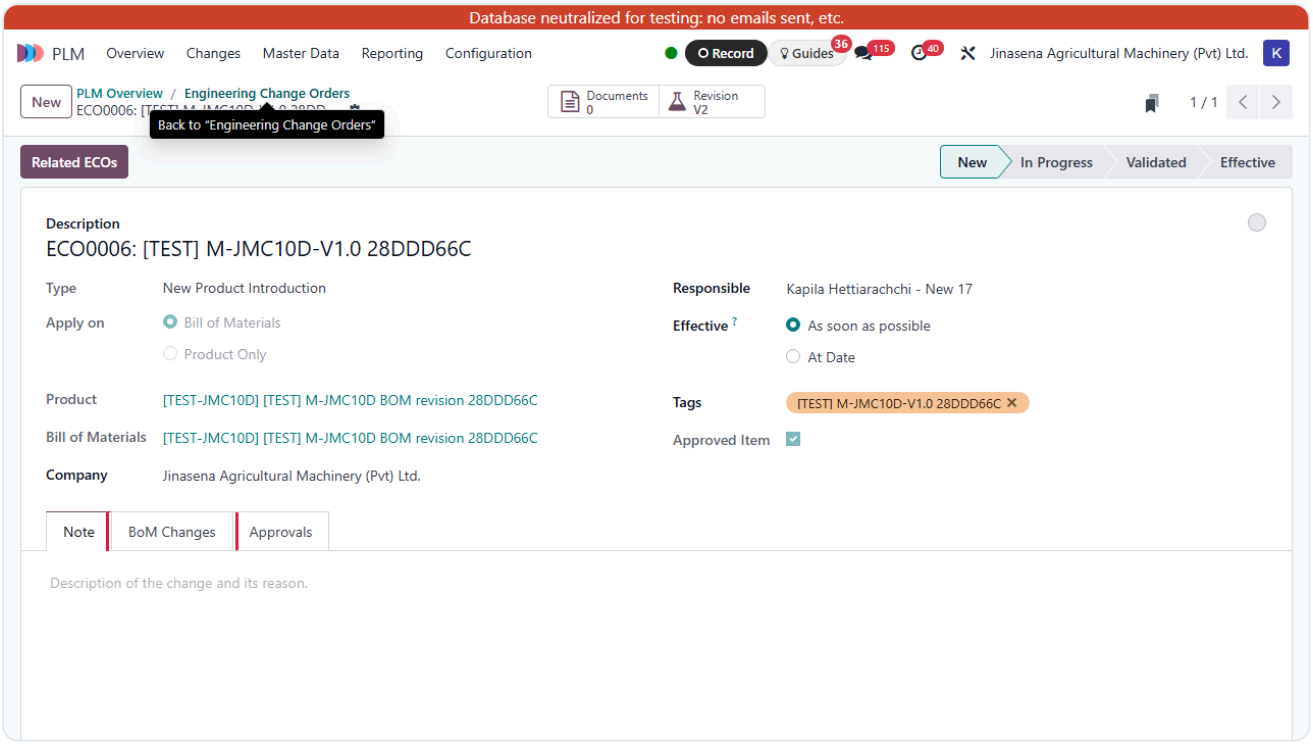
Section D — Review, validate, approve and apply the changes (Steps 14–18)

14 Review the BoM Changes

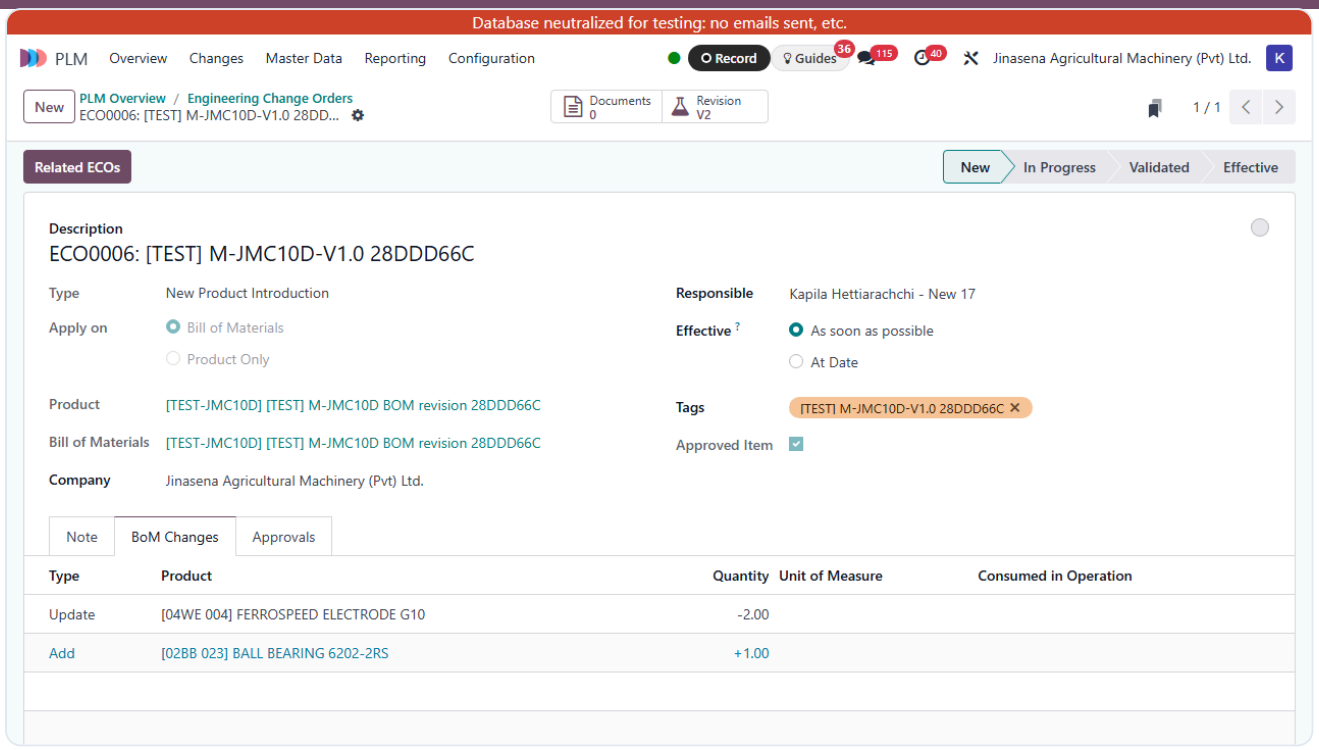
Navigation PLM → Engineering Change Orders → ECO

Action **Click the BoM Changes tab.**

Description Lists every change made in the new version (quantity updates and added components).



ECO — BoM Changes tab



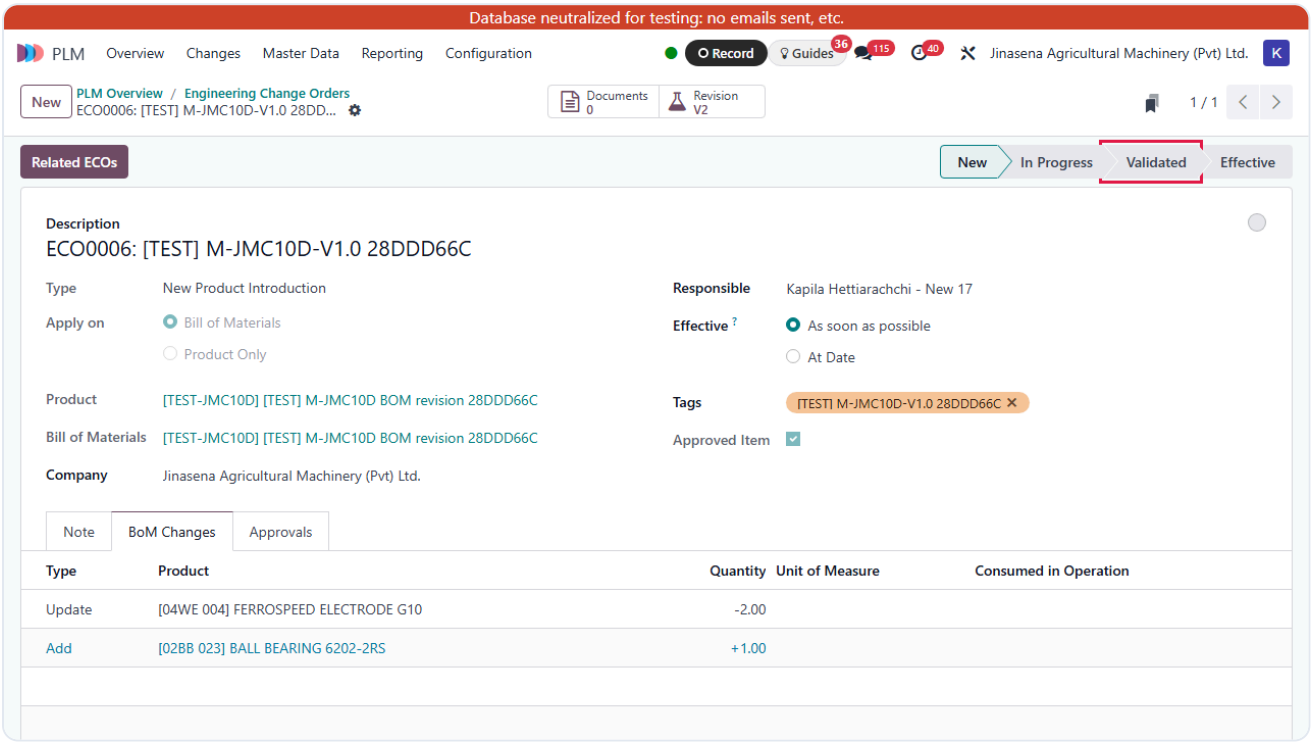
BoM Changes — list of changes

15 Move the ECO to Validated

Navigation PLM → Engineering Change Orders → ECO

Action **Click the Validated stage in the status bar.**

Description The Validated stage requires an approval.



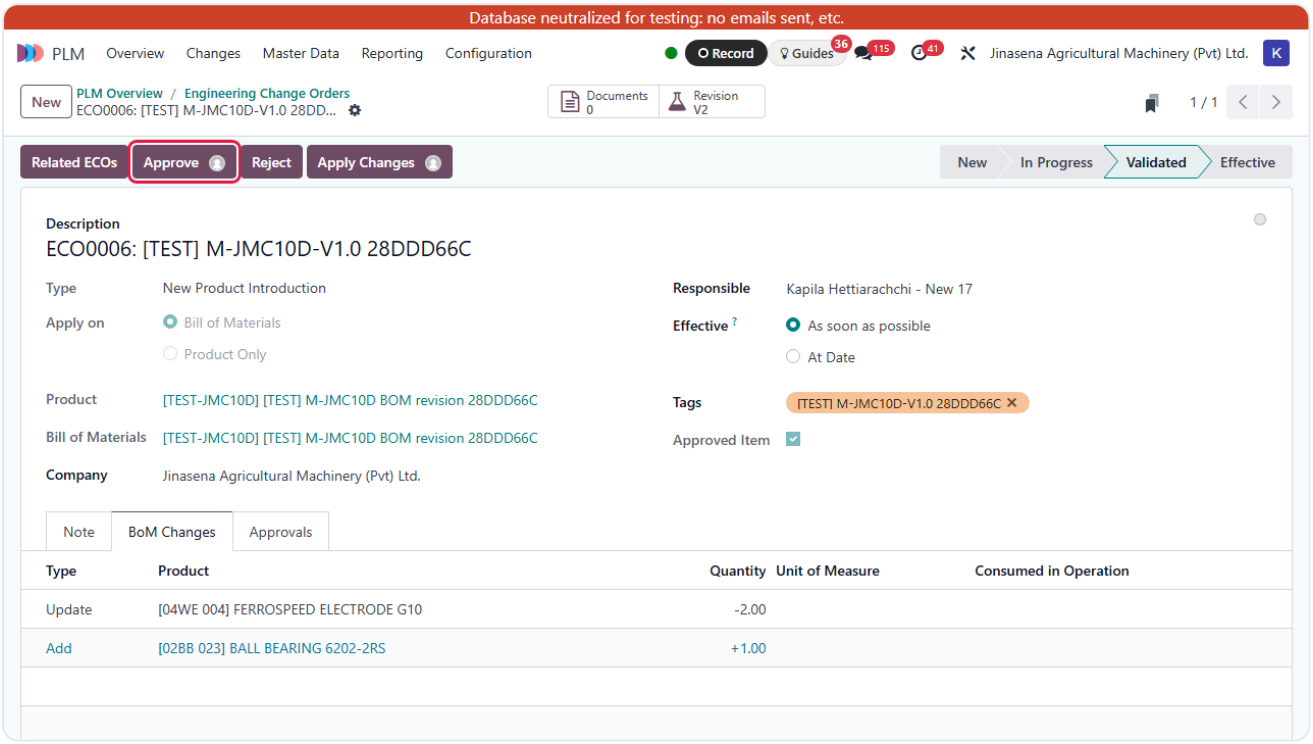
ECO — Validated stage

16 Approve

Navigation PLM → Engineering Change Orders → ECO

Action **Click Approve.**

Description Registers your approval for the ECO.



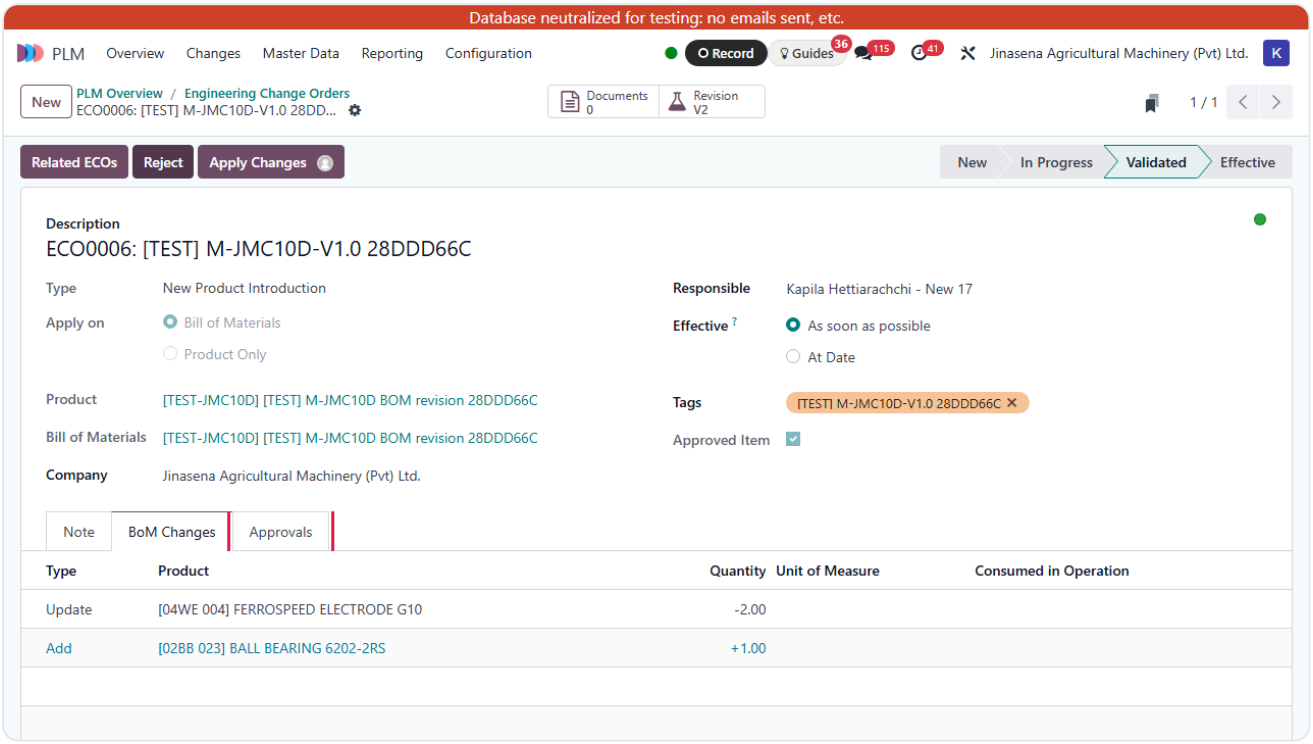
ECO — Approve button

17 See the approval details

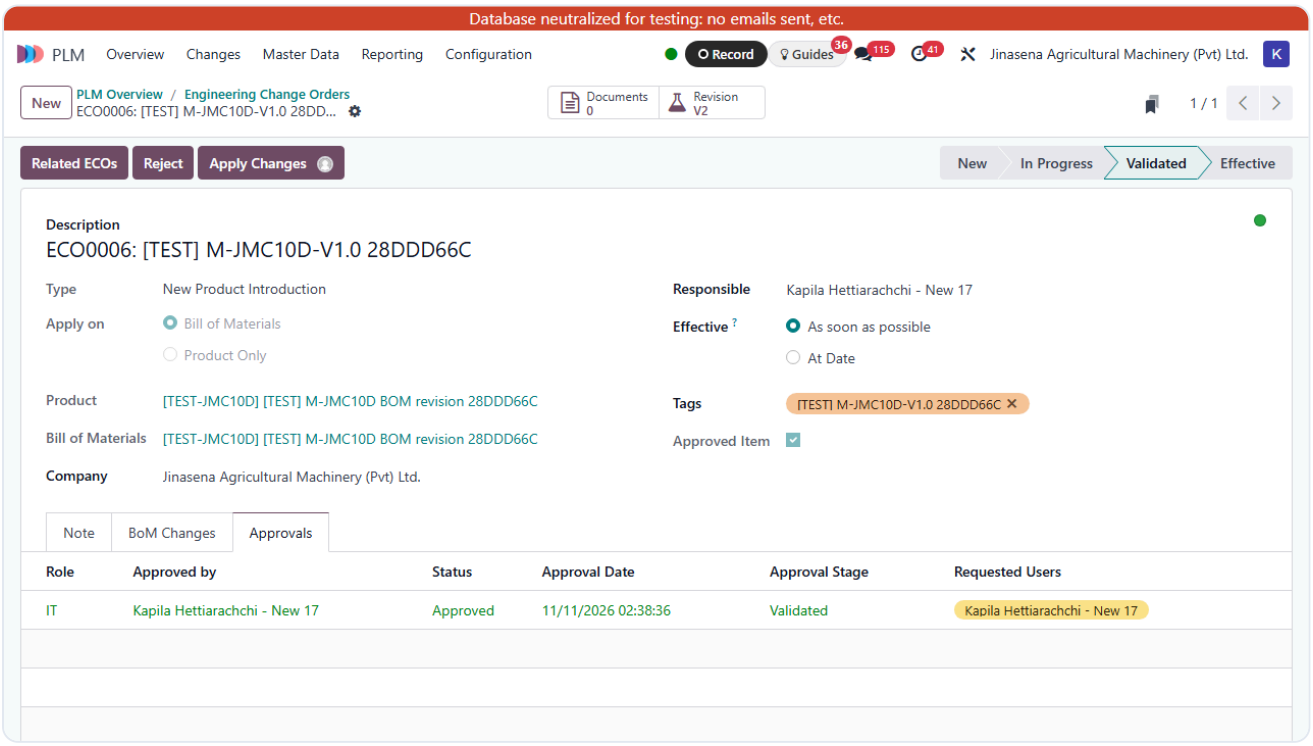
Navigation PLM → Engineering Change Orders → ECO

Action **Click the Approvals tab.**

Description Shows who approved the ECO.



ECO — Approvals tab



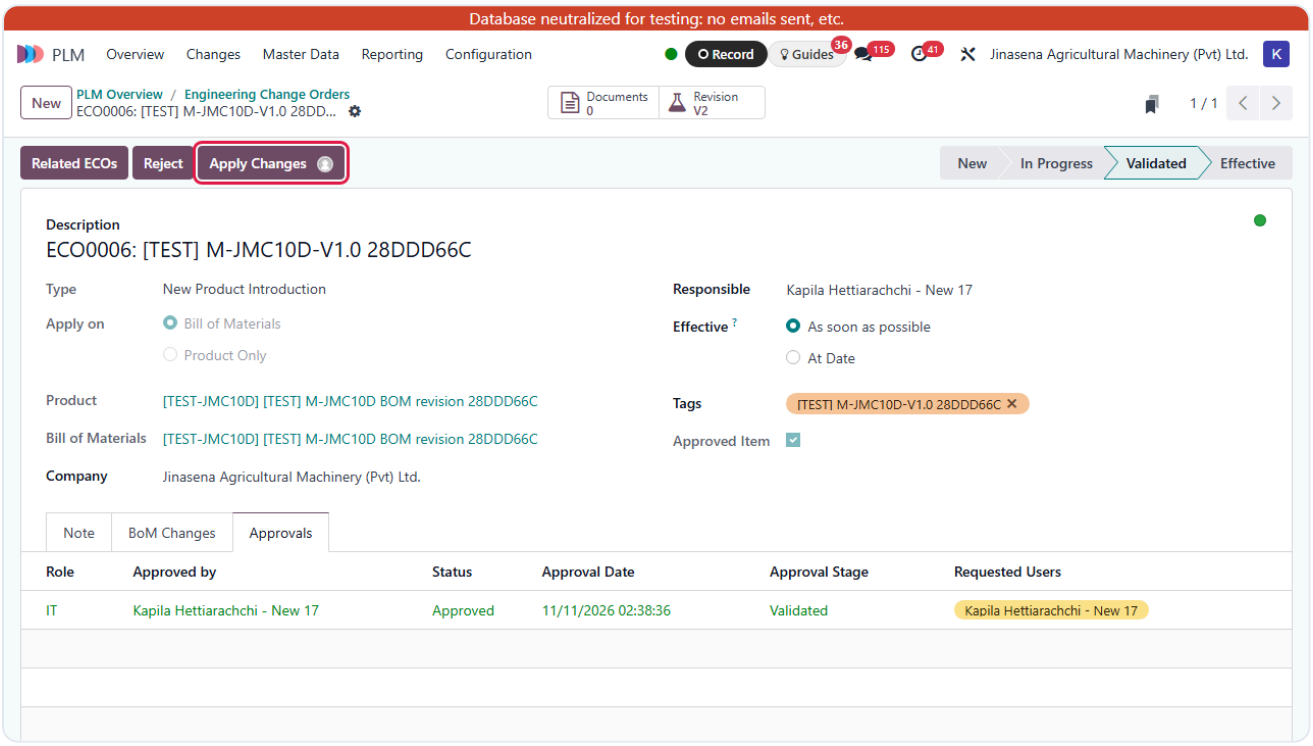
Approvals — approval details

18 Apply the changes

Navigation PLM → Engineering Change Orders → ECO

Action **Click Apply Changes.**

Description Activates the new revision: the new BOM version becomes the active BOM and the previous one is archived.



ECO — Apply Changes button

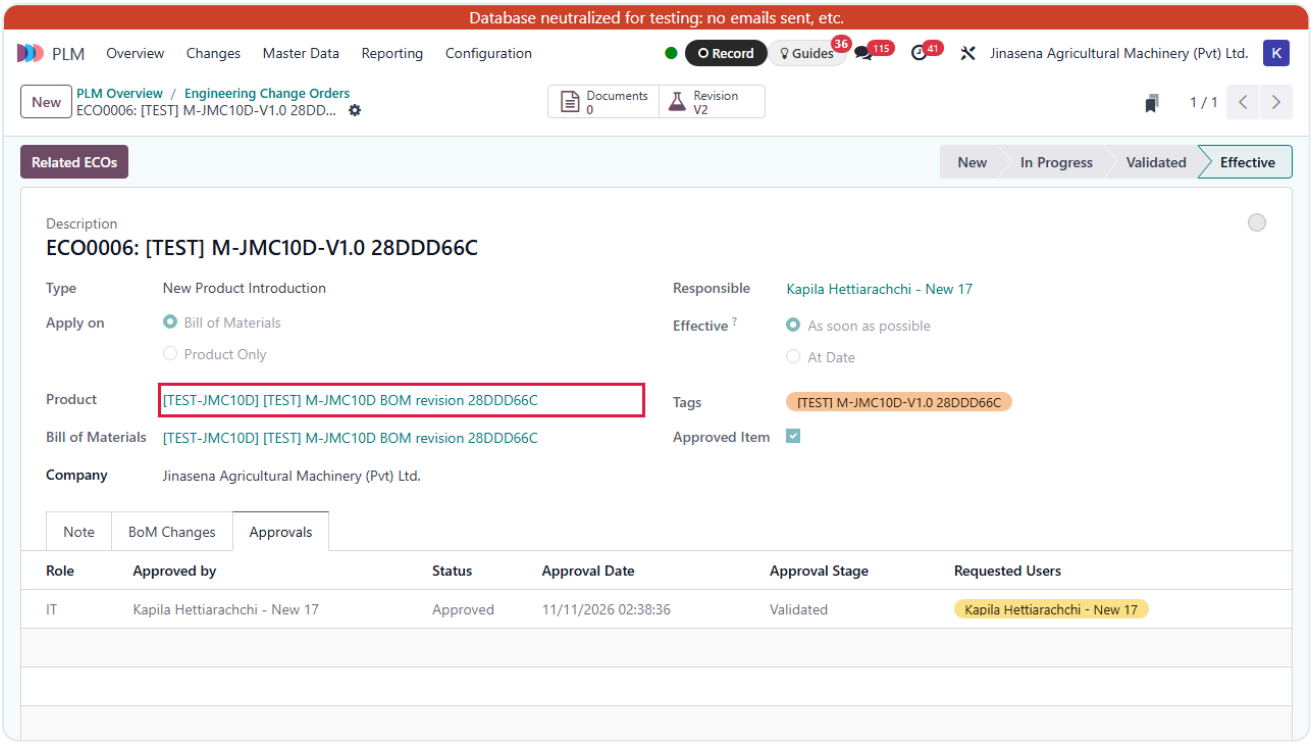
Section E — Check the new BOM version (Steps 19–22)

19 Open the product

Navigation PLM → Engineering Change Orders → ECO

Action **Click the Product link.**

Description Opens the product form so the new BOM version can be checked.



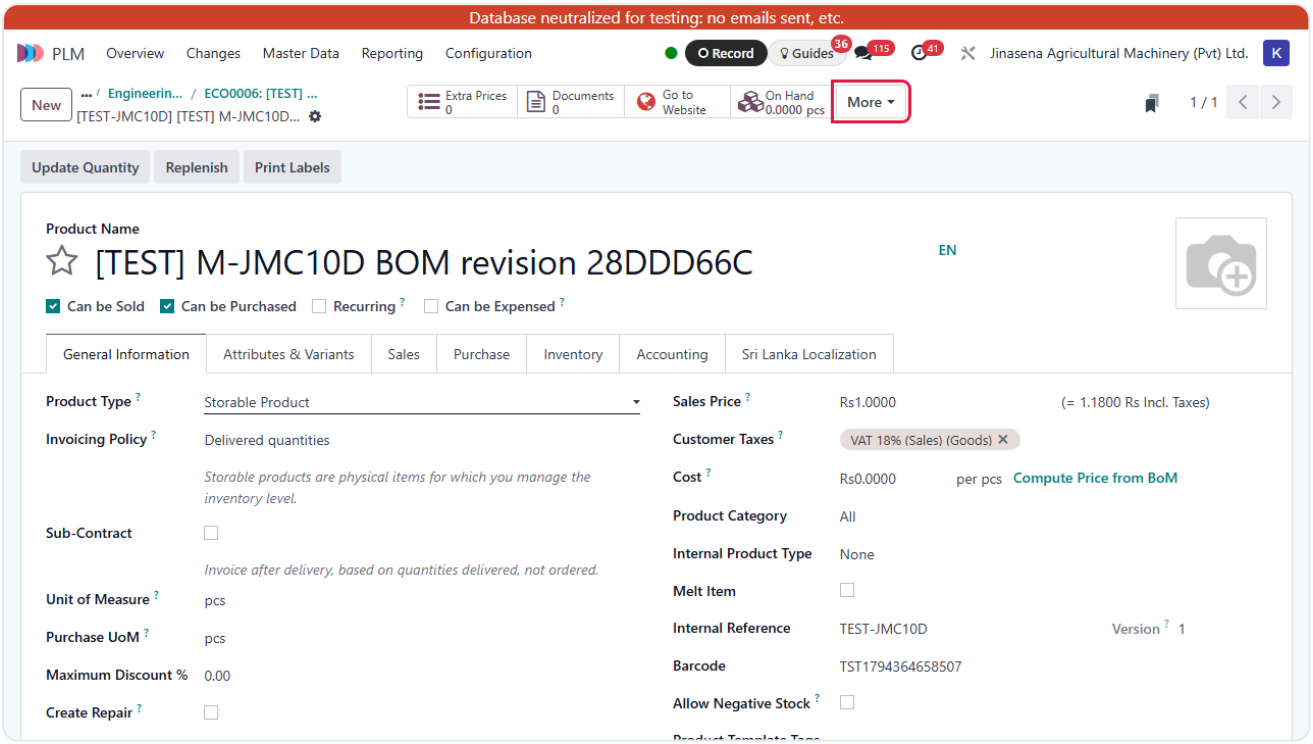
ECO — Product link

20 Open the More menu

Navigation Inventory / Product form

Action **Click More in the button box.**

Description Find the BOM details.



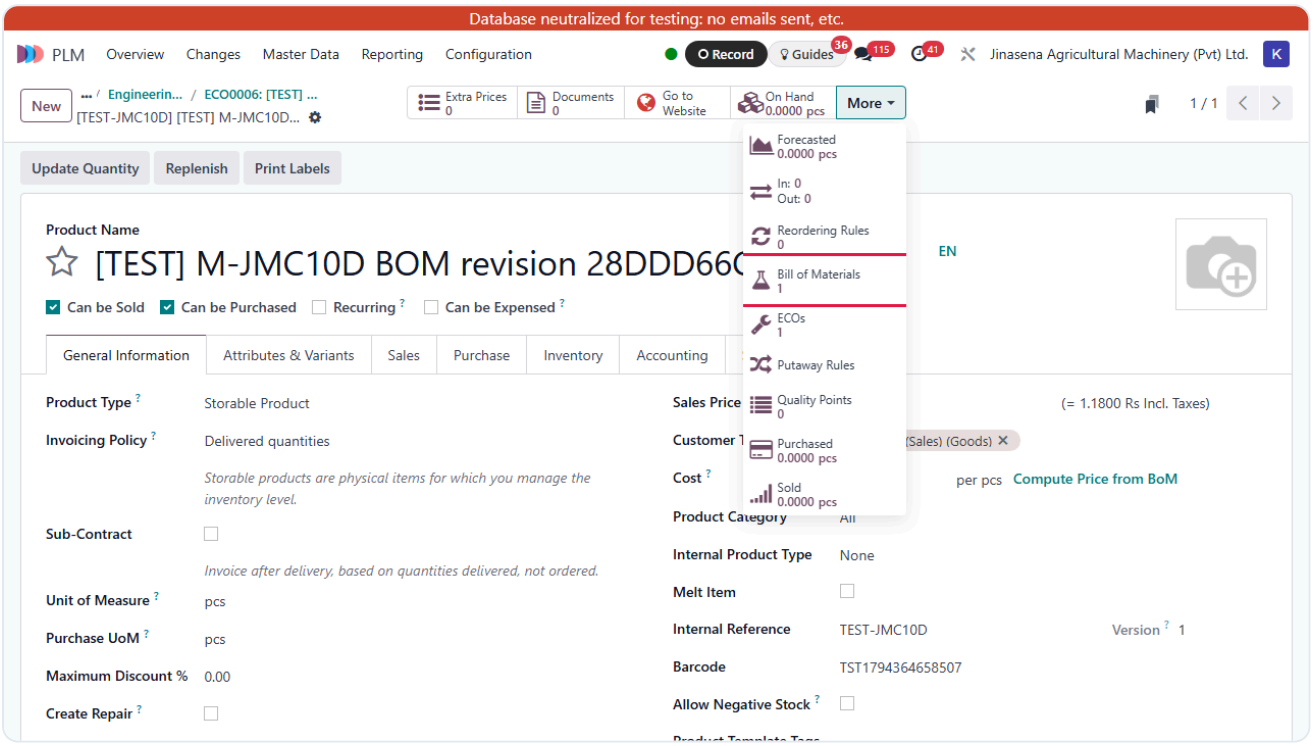
Product — More button

21 Open the Bill of Materials

Navigation Product → More

Action **Click Bill of Materials.**

Description Opens the BOM of the product. When the product has several BOMs a list appears first: click the row to open it.



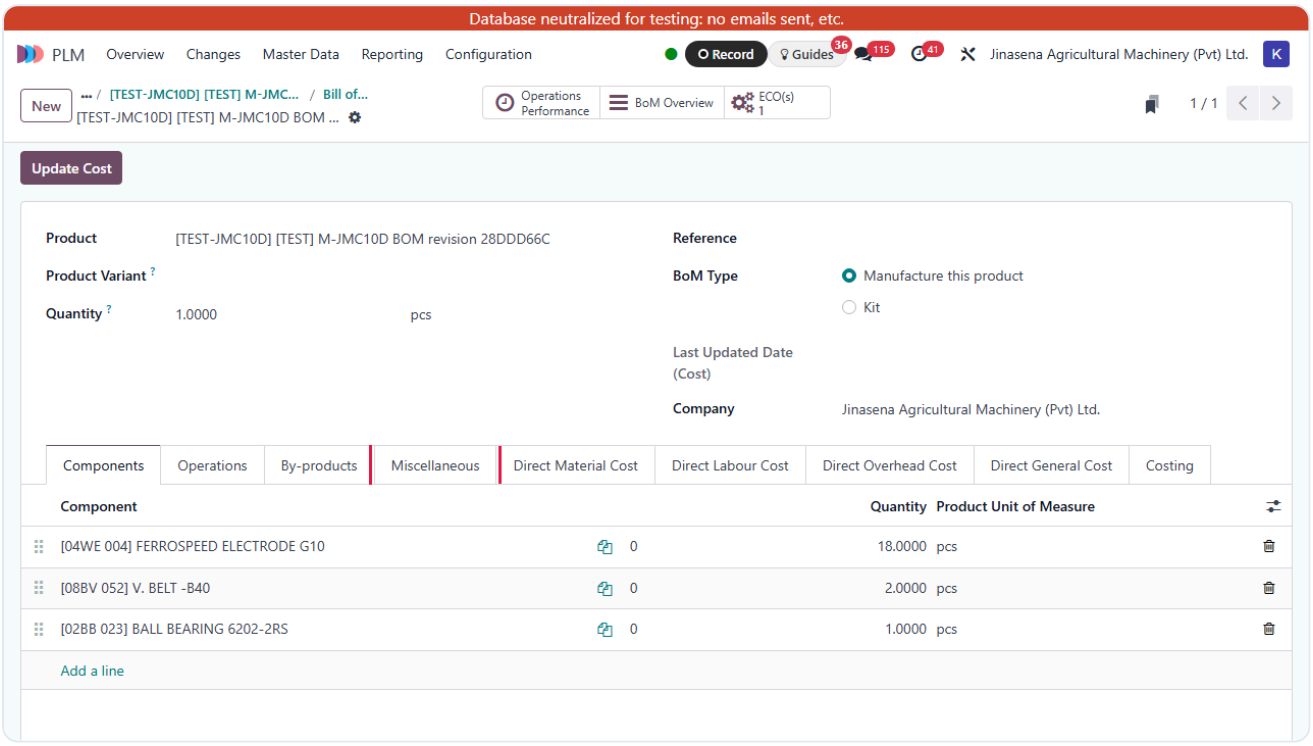
Product — More › Bill of Materials

22 Check the new version

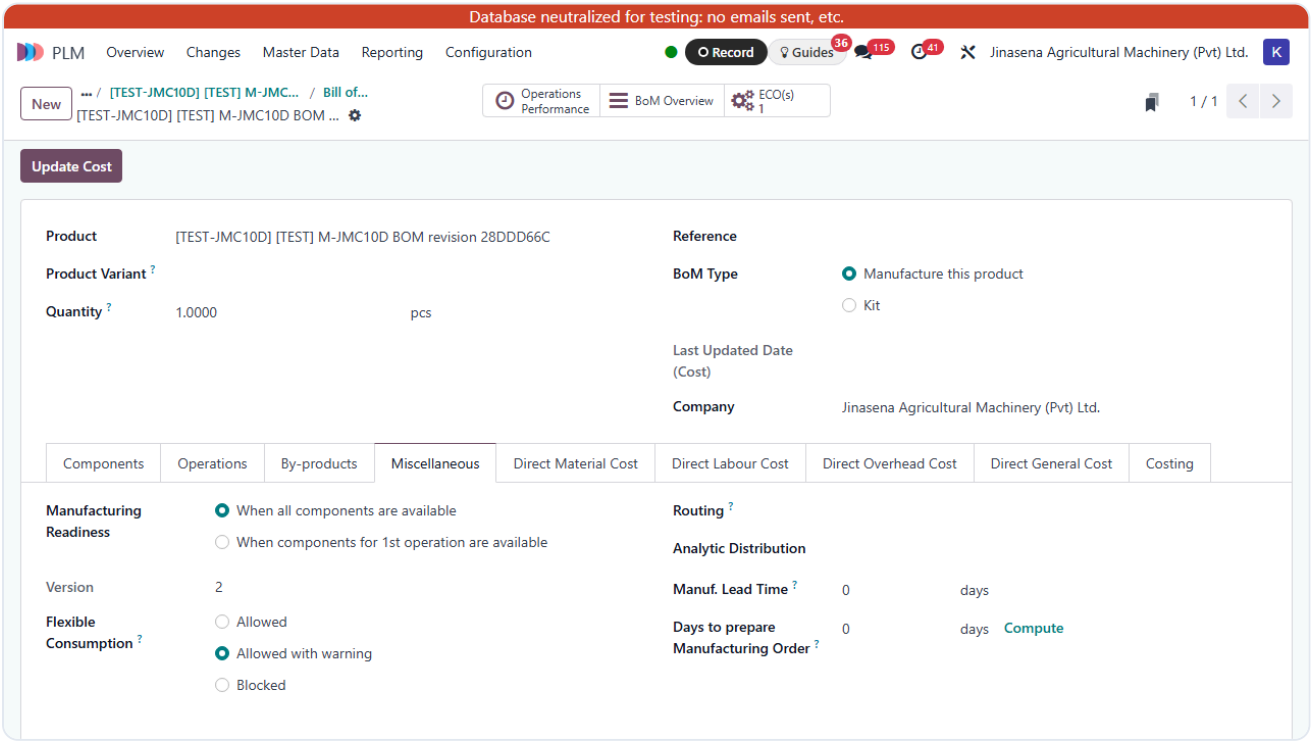
Navigation Bill of Materials

Action **Click the Miscellaneous tab.**

Description The **Version** now shows the new number (2) and the BOM is no longer archived: the new BOM version is active.



Bill of Materials — Miscellaneous tab



Miscellaneous — new Version number

QA notes

- The tour revises the real product `JMC10D`. The automated run does not touch production data: it builds a throw-away `[TEST]` product with a small BOM through RPC (set-up only), flags it as an **Approved Item** (a custom rule — the ECO refuses an unapproved product) and gives it a barcode (required by the product form on this instance). Every documented step is performed through the UI. All created records are `[TEST]`-prefixed and archived afterwards; the screenshots therefore show the test product instead of JMC10D, and the tour's components (e.g. 20 → 18 of the first line) are reproduced as 20 → 18 on the first component of the test BOM.
- The tour has separate steps for typing the ECO name (two steps) and the product; they are merged here into the two steps "Enter the ECO name" and "Select the product". The tour's "Miscellaneous", "Components" and "BoM Changes" clicks are identical in the run.
- After **Apply Changes** the tour clicks the product link, **More**, **Bill of Materials**, then (JMC10D has several BOM versions) a BOM row. The test product has a single active BOM, so Bill of Materials opens it directly and there is no separate "open the row" step; with several BOMs a list appears first.
- The tour's **Back to "ECO Number"** step uses the `b` hotkey; the run clicks the ECO breadcrumb, which does the same.
- The ECO uses type *New Product Introduction* (the instance default); its Validated stage needs one approval, given by the administrator in step 16.
- Each screenshot is captured immediately before the step's action with the target element framed in red (a temporary CSS outline applied to the live element only for the capture); result screenshots have no box. Screenshots come from one clean, passing automated run (admin role).
- Every screenshot shows a red "Database neutralized for testing: no emails sent, etc." banner at the top. This is expected on the test database and is not part of the normal production screen.

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